

Lektura: AI Presentation Producer and Classroom Platform

Project Documentation

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1. Overview

Lektura is a web app that writes presentations. The user answers a few questions (topic, slide count, audience, detail level, optional guidance) and gets an editable deck in about a minute. From there the deck can be edited, narrated in a cloned voice, turned into a video, exported to PowerPoint, or used to make a quiz.

It removes the slow first pass of a presentation: researching, writing slide text and choosing layouts. The user's time goes into improving the deck.

There are three account types: General, Teacher and Student. Teachers run classes and post quizzes. Students join classes and take them.

2. What it does

- Writes a whole deck from a short brief, using live web research when it is available.
- Lays out each slide automatically. The AI chooses the words and the app chooses the layout.
- Lets the user edit text, move and resize elements, change themes and fonts, and draw on slides.
- Writes a speaker script for each slide and reads it in the user's cloned voice.
- Builds a video of the slides with narration.
- Exports editable PowerPoint files.
- Makes quizzes from a deck, printed or taken online.
- Gives teachers classes, where they post quizzes and see student results.

3. How it works

Step	What happens
1. Sign in	The user logs in with an email and password.
2. Brief	A chat-style form asks for the topic, slide count (1 to 7, or let the AI decide), audience, detail level and optional guidance. An unfinished brief is saved as a draft.
3. Research	The AI searches the web for sources and facts on the topic. If the search fails, the deck is still made.
4. Generate	The AI writes every slide in one go.
5. Check	The app checks the slides for quality problems and asks the AI to fix the weakest ones, once.

Step	What happens
6. Edit	The deck opens in the editor. From here all changes are made by hand.
7. Deliver	Present in the browser, export to PowerPoint, make a narrated video, or create a quiz.

A deck's content is generated once, when it is created. For different content, the user starts a new project. Narration, video and quizzes are made separately and leave the slides unchanged.

4. Features

Feature	Description
Deck generation	Turns a short brief into a finished, editable slide deck, backed by web research.
Video presentation generation	Narrates every slide and combines the slides and audio into one downloadable video.
Quiz generation	Makes a quiz from a deck: multiple choice, fill in the blank or true/false, up to 20 items. It can be printed with an answer key or taken online.
Voice cloning	The user records a short sample and the narration is read in their own voice. Premade voices are also available, in six languages.
Face cloning (in development)	A presenter's face will appear as a talking presenter in the narrated video, alongside the cloned voice. This is not built yet.
Slide editor	Edit text in place, move, resize and rotate elements, change layouts, themes and fonts, and draw on slides.
PowerPoint export	Saves the deck as an editable .pptx file.
Classroom	Teachers create classes, post quizzes and see results. Students join with a code and take the quizzes.

5. AI services used

Service	Used for
Anthropic Claude	The first choice for writing decks, web research and narration scripts.
Groq	The backup for decks and narration, and the service that writes quizzes.
Google Gemini	The last backup for decks, research and narration.
Cartesia	Voice cloning and text to speech.
Supabase	Accounts, the database and file storage.

If one AI service is busy, Lektura moves to the next one automatically. Quizzes run on Groq's free tier only.

6. How a deck is generated

- **Research:** the AI gathers sources and facts from the web. This step is skipped when the user asks it to use only their own facts.
- **Writing:** the AI plans each slide (its purpose and key message) and writes the content in the same step.
- **Sources:** claims on the slides are matched to the sources found in research, and each claim is marked as verified or unverified.

- Checking: 18 rules look for problems such as a statistic with no source, too many bullet slides, or a first slide that is not a title slide.
- Repair: up to 4 weak slides are rewritten, once. If the repair fails, the original deck is kept.

The AI follows one of three slide outlines, depending on the goal: informing and training, persuading and selling, or storytelling and engaging.

7. How the video is generated

The video is made in the user's own browser, from the deck as it is. It does not change the slides. It starts when the user presses **Generate Presentation** in the Clone voice window, after choosing a voice.

Before anything else, Lektura checks that the browser can encode video. If it cannot, it stops here, so no voice credits are spent.

Step	What happens
1. Narrate	Each slide's script is sent to Cartesia, which reads it aloud in the chosen voice (cloned or premade), with the chosen language, speed and volume (and emotion, for English). A slide with no script gets 4 seconds of silence.
2. Time	Each slide stays on screen for exactly as long as its narration lasts, so the picture and the sound always line up.
3. Draw	Each slide is drawn off-screen as a 1280 x 720 picture, with its theme, layout, moved elements and drawings. A slide taller than the frame is shrunk to fit, never cut off.
4. Encode	The pictures and the audio are combined into one video file in the browser: MP4 when the browser supports it, otherwise WebM.
5. Save	The video is uploaded to private storage in Supabase. Each deck has one video; making a new one replaces the old one.
6. Watch	The video can be played and downloaded from the same window.

The browser tab must stay visible while the slides are drawn. The user can cancel at any point until the upload begins.

8. Tech stack

Part	Technology
Front end	React 19, TypeScript, Vite
Styling	Tailwind CSS v4
Backend	Supabase (accounts, database, storage). There is no custom server.
AI	Anthropic Claude, Groq, Google Gemini
Voice	Cartesia
Video	Built in the browser with WebCodecs (Mediabunny) and html-to-image
Testing	Vitest